

Vio Commerce x Vipps MobilePay ePayment — solution description

Partner: NewCo AS (Vio Commerce), Norway. **Product:** shoppable media — products sold inside articles, video pages, Vev pages and apps; the merchant's catalogue and orders stay in the merchant's shop (Shopify, WooCommerce) or in Vio. **Vio acts as a platform:** one integration, many merchants (sales units). Test sales unit used for this checklist and by the demo page: MSN 545865 (NewCo AS), through Vio's partner credentials.

Architecture

- `shopcart` (NestJS) owns the Vipps integration: payment creation, status, webhooks, capture / refund / cancel, receipts. `base-api` is the public edge (relays the webhook with its raw body and signature headers; exposes the dashboard routes). `graphql` exposes `InitPaymentVipps` and `GetVippsStatus` to the web SDK. `orders` creates the merchant order and tells `shopcart` when it ships or is cancelled.
- **Credentials per merchant:** *own* (the merchant's keys), *partner* (Vio's partner keys + the merchant's MSN, assigned when Vipps confirms the sales unit), or Vio's own sales unit.
- **Payment creation** (`POST /epayment/v1/payments`): `WALLET`, `WEB_REDIRECT`, `customerInteraction: CUSTOMER_NOT_PRESENT`, `paymentDescription` with the shop and order, `reference VIO-<checkout uuid>` (unique per sales unit, `^[a-zA-Z0-9-]{8,64}$`), `metadata` (checkout, cart, seller), `profile.scope` `name` `address` `email` `phoneNumber`, and for Express `shipping.fixedOptions` (the merchant's rates, brand/type mapped) or `shipping.dynamicOptions` (callback URL + token) with `allowedCountries`. In Express the amount is the goods only; Vipps adds the delivery option the shopper picks.
- **Completion — one path:** webhook, shopper return, status poll and a 10-minute sweep all call the same handler under a database lock; it reads the payment (`GET /payments/{reference}`), acts only on `AUTHORIZED`, creates the order once (idempotent), writes the shopper's details from `userDetails` / `shippingDetails`, sends the receipt (`POST /order-management/v2/ecom/receipts/{reference}`), and captures at once if the merchant chose capture-at-payment.
- **Webhooks:** all eight `epayments.payment.*.v1` events, signature verified (HMAC-SHA256 over `POST\n{path}\n{date};{host};{content-sha256}`, current and previous secret), deduplicated by event id; `authorized` completes, `aborted/expired/terminated/cancelled` close the checkout, `captured/refunded/cancelled` are recorded on the payment's history.
- **Polling:** the SDK polls `GetVippsStatus` on return; the sweep reconciles every open payment every 10 minutes (lost webhooks, closed tabs).
- **Capture:** at payment, at shipment (tracking number or order completion from the merchant's shop or the dashboard) or by hand from the order page; always $\leq$ what is reserved.
- **Refund / cancel:** order cancelled $\rightarrow$ release the reservation or refund the captured amount; partial refund from the order page; a paid reservation whose order could not be created after 24 h is released automatically and the failure logged.
- **Errors:** Vipps' error body is surfaced as `VippsApiError` (status, code, detail, trace id); the shopper sees a plain message ("Betalingen ble avbrutt eller feilet. Vennligst prøv igjen."), the merchant sees the reason in the dashboard; every failing call is logged on one line with endpoint, Vipps' error body (type, title, detail, traceId), the request headers (bearer token and subscription key masked) and the request body.

- **Headers** on every request: `Vipps-System-Name: Vio`, `Vipps-System-Version: 4.0.1` (the shopcart release), `Vipps-System-Plugin-Name: vio-commerce-shopcart`, `Vipps-System-Plugin-Version: 4.0.1`, plus an `Idempotency-Key` on create / capture / refund / cancel.
- **Branding**: Vipps MobilePay's own button component (`<vipps-mobilepay-button>`, from `cdn.vippsmobilepay.com`) on the product, the cart and the checkout.
- **Support**: the order page shows the payment as Vipps sees it (reserved / captured / refunded / released, reference, events) with the actions; support never needs the portal.

Demo

- Test page (Aller Media's Mote & Livsstil, Vio web SDK): <https://mote-livsstil-hub-vio.replit.app/skjonnhet/guider/vio-test-shoppable-favoritter> — open a product and tap the Vipps "buy now" button (Express), or pay with Vipps from the cart, with the Vipps test user.

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